

SPECIMEN

Group 11 Fungicide



# Intrinsic® Brand Fungicide

For use as a drench for soilborne disease control and improved plant health in production ornamentals

**Active Ingredient\*:**

pyraclostrobin: (carbamic acid, [2-[[[1-(4-chlorophenyl)-1H-pyrazol-3-yl]oxy)methyl]phenyl]methoxy-,methyl ester) . . . . . 23.3%

**Other Ingredients:** . . . . . 76.7%

**Total:** . . . . . 100.0%

\* Equivalent to 2.08 pounds of pyraclostrobin per gallon

EPA Reg. No. 7969-355

EPA Est. No.

## KEEP OUT OF REACH OF CHILDREN WARNING/AVISO

Si usted no entiende la etiqueta, busque a alguien para que se la explique a usted en detalle.  
(If you do not understand this label, find someone to explain it to you in detail.)

See inside for complete **First Aid, Precautionary Statements, Directions For Use, Conditions of Sale and Warranty**, and state-specific crop and/or use site restrictions.

**In case of an emergency endangering life or property involving this product, call day or night 1-800-832-HELP (4357).**

## Net Contents:

BASF Agricultural Solutions US LLC  
2 TW Alexander Drive  
Research Triangle Park, NC 27713

**BASF**  
We create chemistry

| FIRST AID                                                                                                                                                                                                                                                                                                                                |                                                                                                                                                                                                                                                                                                                                                                   |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>If swallowed</b>                                                                                                                                                                                                                                                                                                                      | <ul style="list-style-type: none"> <li>• Call a poison control center or doctor immediately for treatment advice.</li> <li>• Have person sip a glass of water if able to swallow.</li> <li>• <b>DO NOT</b> induce vomiting unless told to do so by a poison control center or doctor.</li> <li>• <b>DO NOT</b> give anything to an unconscious person.</li> </ul> |
| <b>If on skin or clothing</b>                                                                                                                                                                                                                                                                                                            | <ul style="list-style-type: none"> <li>• Take off contaminated clothing.</li> <li>• Rinse skin immediately with plenty of water for 15 to 20 minutes.</li> <li>• Call a poison control center or doctor for treatment advice.</li> </ul>                                                                                                                          |
| <b>If in eyes</b>                                                                                                                                                                                                                                                                                                                        | <ul style="list-style-type: none"> <li>• Hold eyes open and rinse slowly and gently with water for 15 to 20 minutes.</li> <li>• Remove contact lenses, if present, after first 5 minutes; then continue rinsing eyes.</li> <li>• Call a poison control center or doctor for advice.</li> </ul>                                                                    |
| <b>If inhaled</b>                                                                                                                                                                                                                                                                                                                        | <ul style="list-style-type: none"> <li>• Move person to fresh air.</li> <li>• If person is not breathing, call 911 or an ambulance; then give artificial respiration, preferably by mouth to mouth, if possible.</li> <li>• Call a poison control center or doctor for further treatment advice.</li> </ul>                                                       |
| HOTLINE NUMBER                                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                                                                                                                   |
| <p>Have the product container or label with you when calling a poison control center or doctor or going for treatment. In case of medical emergency regarding this product, call: BASF Agricultural Solutions US LLC (hereafter "BASF") at 1-800-832-HELP (4357), the National Poison Control Center at 1-800-222-1222, or dial 911.</p> |                                                                                                                                                                                                                                                                                                                                                                   |

## Precautionary Statements

### Hazards to Humans and Domestic Animals

WARNING. May be fatal if swallowed. Causes moderate eye irritation. Avoid contact with skin, eyes or clothing.

### Personal Protective Equipment (PPE)

Some materials that are chemically resistant to this product are listed below. For more options refer to **Category A** on an EPA chemical-resistance category selection chart.

#### Applicators and other handlers must wear:

- Long-sleeved shirt and long pants
- Chemical-resistant gloves made of any waterproof material (such as nitrile, butyl, neoprene and/or barrier laminate)
- Shoes plus socks

Discard clothing and other absorbent materials that have been drenched or heavily contaminated with this product's concentrate. **DO NOT** reuse them.

Follow the manufacturer's instructions for cleaning and maintaining PPE. If no such instructions for washables exist, use detergent and hot water. Keep and wash PPE separately from other laundry.

## USER SAFETY RECOMMENDATIONS

### Users should:

- Wash hands before eating, drinking, chewing gum, using tobacco, or using the toilet.
- Remove clothing/PPE immediately if pesticide gets inside. Then wash thoroughly and put on clean clothing.
- Remove PPE immediately after handling this product. Wash the outside of gloves before removing. As soon as possible, wash thoroughly and change into clean clothing.

## Engineering Controls Statement

When handlers use closed systems, enclosed cabs, or aircraft in a manner that meets the requirements listed in the Worker Protection Standard (WPS) for agricultural pesticides [40 CFR 170.240(d)(4-6)], the handler PPE requirements may be reduced or modified as specified in the WPS.

## Environmental Hazards

This product may contaminate water through drift of spray in wind. This product has a potential for runoff for several months or more after applications. Poorly draining soils and soils with shallow water tables are more prone to produce runoff that contains this product. A level, well-maintained vegetative buffer strip between areas to which this product is applied and surface water features, such as ponds, streams, and springs, will reduce the potential for contamination of water from rainfall runoff. Runoff of this product will be reduced by avoiding applications when rainfall is forecast to occur within 48 hours. Sound erosion control practices will reduce this product's contribution to surface water contamination.

This pesticide is toxic to fish and aquatic invertebrates. Drift and runoff may be hazardous to aquatic organisms in water adjacent to treated areas. **DO NOT** apply directly to water, to areas where surface water is present, or to intertidal areas below the mean high water mark. **DO NOT** contaminate water when disposing of equipment washwaters or rinsate.

Directions For Use

It is a violation of federal law to use this product in a manner inconsistent with its labeling.

**DO NOT** apply this product in a way that will contact workers or other persons, either directly or through drift. Only protected handlers may be in the area during application. For requirements specific to your state or tribe, consult the agency responsible for pesticide regulation.

Unless otherwise directed in supplemental labeling, all applicable directions, restrictions, precautions and **Conditions of Sale and Warranty** are to be followed. This label must be in the user's possession during application.

**AGRICULTURAL USE REQUIREMENTS**

Use this product only in accordance with its labeling and with the Worker Protection Standard, 40 CFR Part 170. This standard contains requirements for the protection of agricultural workers on farms, forests, nurseries, and greenhouses, and handlers of agricultural pesticides. It contains requirements for training, decontamination, notification, and emergency assistance. It also contains specific instructions and exceptions pertaining to the statements on this label about personal protective equipment (PPE), notification of workers, and restricted-entry interval. The requirements in this box only apply to uses of this product that are covered by the Worker Protection Standard.

**DO NOT** enter or allow worker entry into treated areas during the restricted-entry interval (REI) of **12 hours**.

PPE required for early entry to treated areas that is permitted under the Worker Protection Standard and that involves contact with anything that has been treated, such as plants, soil, or water, is:

- Coveralls
- Chemical-resistant gloves made of any waterproof material (such as nitrile, butyl, neoprene, and/or barrier laminate)
- Shoes plus socks

**STORAGE AND DISPOSAL**

**DO NOT** contaminate water, food, or feed by storage or disposal.

**Pesticide Storage**  
Store in original containers only. Keep container closed when not in use. **DO NOT** store near food or feed.

**Pesticide Disposal**  
To avoid wastes, use all material in this container by application according to label directions. If wastes cannot be avoided, offer remaining product to a waste disposal facility or pesticide disposal program (often such programs are run by state or local governments or by industry).

**Container Handling**  
**Nonrefillable Container. DO NOT reuse or refill this container.** Triple rinse or pressure rinse container (or equivalent) promptly after emptying; then offer for recycling, if available, or reconditioning, if appropriate, or puncture and dispose of in a sanitary landfill, or by incineration, or by other procedures approved by state and local authorities.

**Triple rinse containers small enough to shake (5 gallons or less) as follows:** Empty the remaining contents into application equipment or a mix tank and drain for 10 seconds after the flow begins to drip. Fill the container 1/4 full with water and recap. Shake for 10 seconds. Pour rinsate into application equipment or a mix tank, or store rinsate for later use or disposal. Drain for 10 seconds after the flow begins to drip. Repeat this procedure two more times.

**Triple rinse containers too large to shake (capacity > 5 gallons) as follows:** Empty the remaining contents into application equipment or a mix tank. Fill the container 1/4 full with water. Replace and tighten closures. Tip container on its side and roll it back and forth, ensuring at least one complete revolution, for 30 seconds. Stand the container on its end and tip it back and forth several times. Turn the container over onto its other end and tip it back and forth several times. Empty the rinsate into application equipment or a mix tank, or store rinsate for later use or disposal. Repeat this procedure two more times.

**Pressure rinse as follows:** Empty the remaining contents into application equipment or mix tank and continue to drain for 10 seconds after the flow begins to drip. Hold container upside down over application equipment or mix tank, or collect rinsate for later use or disposal. Insert pressure rinsing nozzle in the side of the container and rinse at about 40 PSI for at least 30 seconds. Drain for 10 seconds after the flow begins to drip.

**Container is not safe for food, feed or drinking water.**

## Spills

In case of large-scale spillage regarding this product, call:

- CHEMTREC 1-800-424-9300
- BASF 1-800-832-HELP (4357)

### Steps to be taken in case this material is released or spilled:

- Wear personal protective equipment (see **Precautionary Statements**) and avoid exposure when managing a spill.
- Dike and contain spill with inert material (sand, earth, etc.) and transfer liquid and solid diking material to separate containers for disposal.
- Remove contaminated clothing, and wash affected skin areas with soap and water. Wash clothing before reuse.
- Keep spill out of all sewers and open bodies of water.

---

## Product Information

**Empress® Intrinsic® brand fungicide** is a broad-spectrum fungicide applied as a drench for the control of certain soilborne seedling, root, and crown diseases in production ornamentals, forest and conifer nurseries, and plantations in the following use sites:

- Greenhouses
- Interiorscapes
- Lathhouses and shadehouses
- Outdoor nurseries (container or field)
- Retail nurseries

Applications of **Empress** prior to disease development (preventively) optimize disease control, resulting in improved plant health. Apply **Empress** solo or in tank mixes with other registered fungicides. **DO NOT** exceed the specified application rate or fail to comply with use restrictions listed in this label. Failure to follow directions and precautions on this label can result in injury and/or inferior disease control.

**Empress** is a suspension concentrate (SC) containing the active ingredient pyraclostrobin, a member of the strobilurin class of chemistry derived from a natural antifungal substance. To maximize disease control, apply **Empress** in a regularly scheduled protective drench program and use in a rotation program with other fungicides. **Empress** has good residual activity against target fungi.

---

## Resistance Management

### Mode of Action

Pyraclostrobin, the active ingredient in **Empress** belongs to the group of respiration inhibitors classified as Quinone Outside Inhibitors (QoI) or target site of action **Group 11** fungicides. **Empress** is effective against pathogens resistant to **Non-Group 11** fungicides, such as the dicarboximides, sterol inhibitors, benzimidazoles, or phenylamides. Repeated use of **Group 11** fungicides, including pyraclostrobin, azoxystrobin, fluoxastrobin and trifloxystrobin, can lead to fungal populations that are

resistant to these fungicides. **Empress** will be less effective in controlling diseases where resistant fungi are present.

## Resistance Management Advisory

The following instructions can delay the development of fungicide resistance:

1. **Limited sequential applications** - To maintain performance in ornamental plants, **DO NOT** make more than 2 sequential applications of **Empress**. **DO NOT** alternate **Empress** with other **Group 11** fungicides. Specifically, for diseases caused by *Phytophthora* spp. or *Pythium* spp., alternate with a fungicide(s) of a different mode of action, such as **Orvego® fungicide**, **Stature® SC fungicide**, **Segway® fungicide** or **Subdue® Maxx® fungicide** before reapplying **Empress**.
2. **Tank mixtures** - Tank mixing with fungicides from different target site of action groups that are registered/permitted for the same use and that are effective against the pathogens of concern can delay resistance development. Use at least the minimum labeled rates of each fungicide in the tank mix.
3. **Integrated Pest Management (IPM)** - Integrate **Empress** into an overall disease and pest management program. Follow cultural practices known to reduce disease development. **Empress** can be used in advisory (disease forecasting) programs, which recommend application timing based on environmental factors favorable for disease development.
4. **Monitoring** - Monitor the success of each fungicide in controlling the targeted pathogen and record other factors that can influence fungicide performance and/or disease development. If a **Group 11** target site fungicide, such as **Empress**, appears to be less effective against a pathogen than in previous applications, contact a BASF representative or local expert for further investigation.

---

## Plant Tolerance

**Empress** has been applied to a wide variety of common ornamental plants without observed plant injury. Not all cultivars, varieties or species have been tested for tolerance to **Empress**. Refer to **Table 4** for the list of plants shown to be tolerant to **Empress**.

- Many cultivars within a plant species vary in tolerance to growing conditions and chemical applications. The grower must recognize these differences and test the product accordingly.
- **DO NOT** tank mix combinations with other pesticides or additives that have not been tested for tolerance under local conditions. If tank mixed with another product, always test a small group of representative plants for tolerance prior to large-scale use.
- **DO NOT** tank mix **Empress** with a carbamate or an organophosphate insecticide or a plant growth regulator.
- Local conditions can also influence plant tolerance and may not match those under which BASF has conducted

testing. Always test a small group of representative plants for tolerance to **Empress® Intrinsic® brand fungicide** under local growing conditions prior to large-scale use.

- The use of an adjuvant or surfactant is not necessary when making a drench application.
- **DO NOT** expose plants listed in **Table 5** to drench, spray or drift containing **Empress** as injury can result.

---

### Restrictions and Limitations

- **For outdoor uses, DO NOT** apply more than 1.5 gals of **Empress** product/acre/year, or 4.4 fl ozs per 1,000 sq ft/year. See **Table 2** as a guide for maximum number of applications based on rate and spray volumes.
- **For greenhouse uses, DO NOT** make more than 8 applications of **Empress** per crop per year.
- **DO NOT** apply to plants exhibiting injury (leaf phytotoxicity or plant stunting) produced by prior pesticide applications.
- **DO NOT** tank mix **Empress** with a carbamate or an organophosphate insecticide or a plant growth regulator when making an application.
- **DO NOT** use on crops intended for food or feed use grown in or for agricultural production.
- **DO NOT** use on crops grown in greenhouses for agricultural production or transplants grown in greenhouses for agricultural fields.
- **DO NOT** apply to ornamentals planted in landscape areas, including residential areas, parks, golf courses, cemeteries, roadside plantings or other similar areas.
- **DO NOT** apply by overhead irrigation to fruit and nut trees, blackberries, cranberries, or grapes (see **Table 3**).
- **Resistance Management** - To limit the potential for development of resistance in ornamental plants, **DO NOT** apply more than 2 sequential applications of **Empress**. Rotate to a different mode of action before reapplying **Empress**.
- **DO NOT** expose plants listed in **Table 5** to drench, spray or drift containing **Empress** as injury can result.
- **DO NOT** apply by aircraft.

---

### Mixing Instructions

**Empress** is mixed with water and applied as a soil drench treatment using ground-type application equipment, overhead irrigation equipment, backpack or hand-held equipment.

- Clean spray equipment thoroughly before and after applying this product, particularly if a product with the potential to injure ornamentals was used prior to **Empress**.
- Calibrate drench equipment prior to use.

- Prepare only the amount of drench mix needed for immediate use.
- Maintain maximum constant agitation during application.

### Tank Mix Partners/Components

**Empress** is compatible with most fungicide, insecticide and fertilizer products. If tank mixtures are used, follow rate restrictions, label recommendations and precautions on all labels. Always follow the most restrictive label.

Physical incompatibility, reduced disease control, or plant injury can result from mixing **Empress** with fungicides, herbicides, insecticides, additives, or fertilizers. Always conduct a jar test and test the tank mixes on a small group of representative plants prior to large-scale use.

**DO NOT** tank mix **Empress** with a carbamate or an organophosphate insecticide or a plant growth regulator.

### Additives

**The use of additives or spray adjuvants when making a drench application is not necessary with Empress.**

If additives or spray adjuvants are included, use only surfactants approved for ornamental plants in combination with **Empress**. **DO NOT** use organosilicone-based adjuvants with **Empress** because injury can result on certain ornamental species. Always test the additives and tank mixes on a small group of representative plants prior to large-scale use.

### Compatibility Test (Jar Test)

Before mixing a new combination of products or additives in the spray tank, perform a compatibility test. Begin with a quart-sized jar. Add products in the same order as the **Mixing Order** section. Start with 3.5 cups of water from the intended source at the source temperature. For each dry product, add 2 tsp per pound of product per acre. For each liquid product, add 1 tsp per pint of product per acre.

- Cap the jar and invert 10 cycles between component additions.
- When the components have all been added to the jar, let the solution stand for 15 minutes.
- Evaluate the solution for uniformity and stability. The spray solution should not have free oil on the surface, fine particles that precipitate to the bottom or thick (clabbered) texture. **DO NOT** use any spray solution that could clog spray nozzles.

### Mixing Order

1. **Water** - Fill tank 1/2 to 3/4 full with clean water and start agitation.
2. **Agitation** - Maintain agitation throughout mixing.
3. **Products in PVA bags** - Place any product contained in water-soluble PVA bags into the mixing tank. Wait until bags have fully dissolved and the product is evenly mixed in the spray tank before continuing.
4. **Water-soluble additives** - Including dry and liquid fertilizer, such as ammonium sulfate or urea ammonium.



5. **Water dispersible products** – Including dry flowables, dry wettable granules, suspension concentrates, or suspo-emulsions. **Shake containers well prior to use.** Consult BASF Representatives for additional information regarding agitation and recirculation.
6. **Water soluble products**
7. **Emulsifiable concentrates** – Including oil concentrates or methylated seed oil.
8. **Remaining quantity of water**

If mixture stands for more than 2 hours, thoroughly agitate prior to application.

### Application Directions and Diseases Controlled

Apply **Empress® Intrinsic® brand fungicide** as a preventive drench application for control of specific soilborne pathogens causing root rot, stem rot, and crown rot in ornamental plants. Follow directions for use found in **Tables 1, 2 and 3.**

**Shake containers well prior to use.** Consult BASF Representatives for additional information regarding agitation and recirculation.

### Drench Application Instructions

- Drench the soil with a solution of **Empress** using **Tables 1, 2 and 3.** Thorough saturation and wetting of root zone, crown and base of the plant, and surrounding growth media is necessary for optimum control.
- **Drench - DO NOT** apply to dry soil media. Applications are best made when soil media has good moisture but not saturated. Use enough solution to wet the root zone of the plant. To facilitate product uptake and maximize fungicide contact, limit watering for several hours after application except to the extent needed to move the fungicide into the soil.
- **For Seedling Ornamentals - DO NOT** apply **Empress** until both roots and a first true leaf are present. Cuttings should be rooted prior to use of **Empress.**
- **DO NOT** use **Empress** after symptoms of soilborne disease have become evident. Control may not be satisfactory.
- Repeat applications as needed 7 to 28 days after the initial application.
- **Resistance Management** - To limit the potential for development of resistance in ornamental plants, **DO NOT** apply more than 2 sequential applications of **Empress.** Rotate to a different mode of action before reapplying **Empress.**
- **Empress** can be applied by ground-type application equipment, overhead irrigation (see **Table 3**), backpack or hand-held equipment.
- See **Table 2** for instructions on total number of applications per rate and drench volume for outdoor use.

**Table 1. Empress Mixing Rate and Diseases Controlled**

| Crown, Root, and Soilborne Pathogens                                                               | Product Use Rate per 100 Gal (fl ozs) | Production Unit                                                                                                                                                                                                     |
|----------------------------------------------------------------------------------------------------|---------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <i>Fusarium</i> spp.<br><i>Phytophthora</i> spp.<br><i>Pythium</i> spp.<br><i>Rhizoctonia</i> spp. | 1.0 to 3.0                            | In Propagation:<br>Rooted cuttings<br>Seedlings<br>Plugs                                                                                                                                                            |
|                                                                                                    | 2.0 to 6.0                            | In Production:<br>Greenhouse/<br>lathhouse grown<br>ornamentals, annuals, or perennial plants (e.g. bedding plants, pot crops, and foliage)<br><br>Container and field grown herbaceous and woody ornamental plants |

**Table 2. Empress Maximum Applications by Rate and Volume**

| Production Unit                                                                   | Maximum Gal/A | Maximum Rate per 100 Gal/ Application* (fl ozs) | Maximum Rate per Application (fl ozs/A) |
|-----------------------------------------------------------------------------------|---------------|-------------------------------------------------|-----------------------------------------|
| Rooted cuttings, seedlings, and plugs in propagation                              | 400           | 3                                               | 12                                      |
| Herbaceous and woody plants in greenhouse, nursery container, or field production | 1000          | 6                                               | 60                                      |

\* **DO NOT** apply more than 1.5 gals of **Empress** product/acre/year, or 4.4 fl ozs per 1000 sq ft/year. **DO NOT** exceed the spray volume maximum per acre and the maximum rate mixed per 100 gal when making applications.

Resistance management dictates a maximum of two sequential applications then rotate to a **non-group 11** fungicide with a different MOA.

**Table 3. Drench Applications To Fruit and Nut Trees, Blackberries, Cranberries, or Grapes**

**DO NOT** use on crops intended for food or feed use grown in or for agricultural production.

**DO NOT** apply using overhead irrigation or other equipment that delivers application to foliage. **Target drench application to container surface and soil only.**

| Common Name         | Scientific Name                         |
|---------------------|-----------------------------------------|
| Almond              | <i>Prunus dulcis</i>                    |
| Apple               | <i>Malus</i> spp.                       |
| Apricot             | <i>Prunus armeniaca</i>                 |
| Blackberry          | <i>Vaccinium myrtillus</i>              |
| Cherry              | <i>Prunus avium</i> , <i>P. cerasus</i> |
| Chestnut, American  | <i>Castanea pumila</i>                  |
| Citrus              | <i>Citrus</i> spp.                      |
| Cranberry, American | <i>Vaccinium macrocarpon</i>            |
| Grape, European*    | <i>Vitis vinifera</i>                   |
| Nectarine           | <i>Prunus persica</i>                   |
| Peach               | <i>Prunus persica</i>                   |
| Pear                | <i>Pyrus</i> spp.                       |
| Pistachio           | <i>Pistacia vera</i>                    |
| Plum                | <i>Prunus domestica</i>                 |
| Walnut, black       | <i>Juglans nigra</i>                    |
| Walnut, common      | <i>Juglans regia</i>                    |

\* **DO NOT** expose these Grape species or varieties to **Empress® Intrinsic® brand fungicide**: Concord, Fredonia, Niagara, Noiret, (NY73.0136.17), Rougeon, Steuben and Worden.

## Overhead Irrigation Systems

**DO NOT** apply when wind speed favors drift beyond the area intended for treatment.

Applications made through irrigation systems must deliver rates of **Empress** consistent with the rates specified in **Tables 1** and **2** of this label.

**DO NOT** apply by overhead irrigation to fruit and nut trees, blackberries, cranberries, or grapes (see **Table 3**).

Apply this product through sprinkler or drip irrigation systems, including center pivot, lateral move, end tow, side [wheel] roll, traveler, big gun, solid set, or hand move irrigation systems.

Plant injury, lack of effectiveness, or illegal pesticide residues in the crop can result from nonuniform treated water. Thorough **soil drench** is required for good control. Maintain good agitation during the entire application period. If you have questions about calibration, contact a state extension service specialist, equipment manufacturers or other experts.

- The system must contain a functional check valve, vacuum-relief valve, and low-pressure drain appropriately located on the irrigation pipeline to prevent water-source contamination from backflow.

- The pesticide injection pipeline must contain a functional, automatic quick-closing check valve to prevent the flow of fluid back toward the injection pump.
- The pesticide injection pipeline must also contain a functional, normally closed solenoid-operated valve located on the intake side of the injection pump and connected to the system interlock to prevent fluid from being withdrawn from the supply tank when the irrigation system is either automatically or manually shut down.
- The system must contain functional interlocking controls to automatically shut off the pesticide injection pump when the water pump motor stops.
- The irrigation line or water pump must include a functional pressure switch, which will stop the water pump motor when the water pressure decreases to the point where pesticide distribution is adversely affected.
- Systems must use a metering pump, such as a positive displacement injection pump (e.g. diaphragm pump) effectively designed and constructed of materials that are compatible with pesticides and capable of being fitted with a system interlock.
- Allow sufficient time for pesticide to be flushed through all lines and all nozzles before turning off irrigation water. A person knowledgeable of the chemigation system and responsible for its operation, or under the supervision of the responsible person, shall shut the system down and make necessary adjustments should the need arise.

**DO NOT** connect an irrigation system (including greenhouse systems) used for pesticide application to a public water system unless the pesticide label-prescribed safety devices for public water systems are in place.

## Specific Instructions for Public Water Systems

- Public water system means a system for the provision to the public of piped water for human consumption if such system has at least 15 service connections or regularly serves an average of at least 25 individuals daily at least 60 days out of the year.
- Chemigation systems connected to public water systems must contain a functional reduced-pressure zone backflow preventer (RPZ) or the functional equivalent in the water supply line upstream from the point of pesticide introduction. As an option to the RPZ, discharge the water from the public water system into a reservoir tank prior to pesticide introduction. There shall be a complete physical break (air gap) between the outlet end of the fill pipe and the top or overflow rim of the reservoir tank of at least twice the inside diameter of the fill pipe.
- The pesticide injection pipeline must contain a functional, automatic quick-closing check valve to prevent the flow of fluid back toward the injection pump.
- The pesticide injection pipeline must contain a functional, normally closed solenoid-operated valve located on the intake side of the injection pump and connected to the system interlock to prevent fluid from being withdrawn from the supply tank when the irrigation system is either automatically or manually shut down.
- The system must contain functional interlocking controls to automatically shut off the pesticide injection pump when the water pump motor stops, or in cases where

there is no water pump, when the water pressure decreases to the point where pesticide distribution is adversely affected.

- Systems must use a metering pump, such as a positive displacement injection pump (e.g. diaphragm pump) effectively designed and constructed of materials that are compatible with pesticides and capable of being fitted with a system interlock.
- **DO NOT** apply when wind speed favors drift beyond the area intended for treatment.

**Table 4. Empress® Intrinsic® brand fungicide Tolerant Plant Species**

Plants in this table have been found to be tolerant to **Empress** when applied according to the use instructions stated in this label.

| Common Name          | Scientific Name                |
|----------------------|--------------------------------|
| African violet       | <i>Saintpaulia ionantha</i>    |
| Ajuga                | <i>Ajuga reptans</i>           |
| Almond, nonbearing   | <i>Prunus dulcis</i>           |
| Aloe vera            | <i>Aloe vera</i>               |
| Apple, nonbearing    | <i>Malus</i> spp.              |
| Apricot, nonbearing  | <i>Prunus armeniaca</i>        |
| Arborvitae           | <i>Thuja</i> spp.              |
| Ardisia              | <i>Ardisia</i> spp.            |
| Arrowwood            | <i>Viburnum dentatum</i>       |
| Ash, red             | <i>Fraxinus pennsylvanica</i>  |
| Asian trache         | <i>Lospermum</i> spp.          |
| Asparagus fern       | <i>Asparagus densiflorus</i>   |
| Astilbe              | <i>Astilbe</i> spp.            |
| Aucuba               | <i>Aucuba japonica</i>         |
| Avens                | <i>Geum chiloense</i>          |
| Azalea               | <i>Rhododendron</i> spp.       |
| Baby's breath        | <i>Gypsophila repens</i>       |
| Bachelor button      | <i>Centaurea montana</i>       |
| Balloon flower       | <i>Platycodon grandiflorus</i> |
| Barbados lily        | <i>Hippeastrum vittatum</i>    |
| Barberry, Japanese   | <i>Berberis thunbergii</i>     |
| Basket-of-gold       | <i>Aurinia saxatilis</i>       |
| Bayberry, wax myrtle | <i>Myrica cerifera</i>         |
| Bee balm             | <i>Monarda didyma</i>          |
| Begonia*             | <i>Begonia</i> spp.            |
| Bellflower           | <i>Campanula glomerata</i>     |
| Blackberry           | <i>Vaccinium myrtillus</i>     |
| Black-eyed Susan     | <i>Rudbeckia</i> spp.          |
| Blanket flower       | <i>Gaillardia grandiflora</i>  |
| Blue lily turf       | <i>Liriope</i> spp.            |

(continued)

**Table 4. Empress® Intrinsic® brand fungicide Tolerant Plant Species** (continued)

| Common Name                            | Scientific Name                                            |
|----------------------------------------|------------------------------------------------------------|
| Boxwood, Japanese, common              | <i>Buxus</i> - <i>B. japonica</i> , <i>B. sempervirens</i> |
| Brachycome, blue                       | <i>Brachycome</i> spp.                                     |
| Bridal wreath                          | <i>Spiraea vanhouttei</i>                                  |
| Butterfly bush                         | <i>Buddleia</i> spp.                                       |
| Caladium                               | <i>Caladium</i> spp.                                       |
| Camellia, Japanese                     | <i>Camellia japonica</i>                                   |
| Canna                                  | <i>Canna</i> x <i>generalis</i>                            |
| Carnation                              | <i>Dianthus caryophyllus</i>                               |
| Cedar, Japanese                        | <i>Cryptomeria japonica</i>                                |
| Chamaecyparis                          | <i>Chamaecyparis pisifera</i>                              |
| Cherry, nonbearing                     | <i>Prunus avium</i> , <i>P. cerasus</i>                    |
| Cherry, flowering, Kwanzan             | <i>Prunus serrulata</i> 'Kwanzan'                          |
| Cherry, flowering, Mt. Fuji (Shirotae) | <i>Prunus serrulata</i> 'Mt. Fuji' (Shirotae)              |
| Chestnut, American                     | <i>Castanea dentata</i>                                    |
| China, rose                            | <i>Hibiscus</i> spp.                                       |
| Chinquapin                             | <i>Castanea pumila</i>                                     |
| Chrysanthemum                          | <i>Chrysanthemum</i> spp.                                  |
| Citrus, nonbearing                     | <i>Citrus</i> spp.                                         |
| Columbine                              | <i>Aquilegia</i> spp.                                      |
| Cone flower                            | <i>Rudbeckia hirta</i>                                     |
| Coral bells                            | <i>Heuchera</i> spp.                                       |
| Cortaderia                             | <i>Cortaderia</i> spp.                                     |
| Cotoneaster, cranberry                 | <i>Cotoneaster apiculatus</i>                              |
| Crabapple                              | <i>Malus</i> spp.                                          |
| Cranberry, American                    | <i>Vaccinium macrocarpon</i>                               |
| Crape myrtle                           | <i>Lagerstroemia indica</i>                                |
| Cryptomeria                            | <i>Cryptomeria</i> spp.                                    |
| Cupid's dart                           | <i>Catananche cerulea</i>                                  |
| Cyclamen                               | <i>Cyclamen</i> spp.                                       |
| Daffodil                               | <i>Narcissus pseudonarcissus</i>                           |
| Dahlia                                 | <i>Dahlia</i> spp.                                         |
| Daylily                                | <i>Hemerocallis</i> spp.                                   |
| Deutzia                                | <i>Deutzia</i> spp.                                        |
| Dietes                                 | <i>Dietes vegeta</i>                                       |
| Dogwood                                | <i>Cornus</i> spp.                                         |
| Douglas fir                            | <i>Pseudotsuga</i> spp.                                    |
| Dusty Miller                           | <i>Centaurea cineraria</i>                                 |
| Enkianthus, redvein                    | <i>Enkianthus campanulatus</i>                             |
| Echinacea                              | <i>Echinacea purpurea</i>                                  |
| Elaeagnus, Russian olive               | <i>Elaeagnus angustifolia</i>                              |

(continued)



**Table 4. Empress® Intrinsic® brand fungicide  
Tolerant Plant Species** (continued)

| Common Name                      | Scientific Name                                                              |
|----------------------------------|------------------------------------------------------------------------------|
| Elder, water                     | <i>Sambucus</i> spp.                                                         |
| Euonymus                         | <i>Euonymus alata</i>                                                        |
| Fern, Kimberly Queen             | <i>Nephrolepis obliterated</i>                                               |
| Fern, wood                       | <i>Dryopteris</i> spp.                                                       |
| Forsythia                        | <i>Forsythia</i> spp.                                                        |
| Fountain grass                   | <i>Pennisetum alopecuroides</i>                                              |
| Foxglove                         | <i>Digitalis</i> spp.                                                        |
| Gardenia                         | <i>Gardenia jasminoides</i>                                                  |
| Gayfeather                       | <i>Liatris</i> spp.                                                          |
| Gazania                          | <i>Gazania</i> spp.                                                          |
| Geranium                         | <i>Pelargonium</i> spp.                                                      |
| Gerbera                          | <i>Gerbera</i> spp.                                                          |
| Germander, Wall                  | <i>Teucrium chamaedrys</i>                                                   |
| Gladiolus                        | <i>Gladiolus</i> spp.                                                        |
| Globe thistle                    | <i>Echinops ritro</i>                                                        |
| Goldbell tree, Chinese           | <i>Forsythia viridissima</i>                                                 |
| Grape, European, nonbearing      | <i>Vitis vinifera</i>                                                        |
| Hawthorn, Indian                 | <i>Rhaphiolepis</i> spp.                                                     |
| Hazel                            | <i>Corylopsis</i> spp.                                                       |
| Heavenly bamboo                  | <i>Nandina domestica</i>                                                     |
| Hemlock, Canada                  | <i>Tsuga Canadensis</i>                                                      |
| Holly, Chinese, Japanese, Yaupon | <i>Ilex</i> - <i>I. cornuta</i> ,<br><i>I. crenata</i> , <i>I. vomitoria</i> |
| Hosta                            | <i>Hosta</i> spp.                                                            |
| Hydrangea                        | <i>Hydrangea</i> spp.                                                        |
| Impatiens*, New Guinea, balsam   | <i>Impatiens</i> spp.                                                        |
| Iris                             | <i>Iris</i> spp.                                                             |
| Ivy, common, California, English | <i>Hedera</i> spp.                                                           |
| Jasmine, star                    | <i>Trachelospermum jasminoides</i>                                           |
| Jessamine                        | <i>Gelsemium sempervirens</i>                                                |
| Juniper, creeping, Chinese       | <i>Juniperus</i> -<br><i>J. horizontalis</i> ,<br><i>J. chinensis</i>        |
| Lamb's ear                       | <i>Stachys byzantina</i>                                                     |
| Lantana                          | <i>Lantana montevidensis</i>                                                 |
| Larkspur                         | <i>Delphinium elatum</i>                                                     |
| Leopard's bane                   | <i>Doronicum cordatum</i>                                                    |
| Leucophyllum                     | <i>Leucophyllum</i> spp.                                                     |
| Lilac, common                    | <i>Syringa</i> spp.                                                          |
| Lily                             | <i>Lilium</i> spp.                                                           |

(continued)

**Table 4. Empress® Intrinsic® brand fungicide  
Tolerant Plant Species** (continued)

| Common Name                                         | Scientific Name                                                                                                                                                       |
|-----------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Liriope, variegated                                 | <i>Liriope muscari variegata</i>                                                                                                                                      |
| Lisianthus                                          | <i>Eustoma grandiflora</i>                                                                                                                                            |
| Lobelia                                             | <i>Lobelia</i> spp.                                                                                                                                                   |
| Loropetalum                                         | <i>Loropetalum chinense</i>                                                                                                                                           |
| Lupine                                              | <i>Lupinus</i> spp.                                                                                                                                                   |
| Magnolia, star, saucer                              | <i>Magnolia</i> -<br><i>M. stellata</i> ,<br><i>M. soulangiana</i>                                                                                                    |
| Maidenhair tree                                     | <i>Ginkgo biloba</i>                                                                                                                                                  |
| Mandevilla                                          | <i>Mandevilla</i> spp.                                                                                                                                                |
| Maple, amur, Japanese, Norway, sugar, soft, negundo | <i>Acer</i> spp. -<br><i>A. ginnala</i> ,<br><i>A. palmatum</i> ,<br><i>A. platanoides</i> ,<br><i>A. saccharum</i> ,<br><i>A. saccharinum</i> ,<br><i>A. negundo</i> |
| Marigold                                            | <i>Tagetes</i> spp.                                                                                                                                                   |
| Maudlin, blue                                       | <i>Ageratum houstonianum</i>                                                                                                                                          |
| Meadow sage                                         | <i>Salvia x superba</i>                                                                                                                                               |
| Monkey grass                                        | <i>Ophiopogon japonicus</i>                                                                                                                                           |
| Morningglory                                        | <i>Ipomoea</i> spp.                                                                                                                                                   |
| Moss, rose                                          | <i>Portulaca grandiflora</i>                                                                                                                                          |
| Mountain laurel                                     | <i>Kalmia latifolia</i>                                                                                                                                               |
| Myrica cerifera                                     | <i>Myrica cerifera</i>                                                                                                                                                |
| Myrtle                                              | <i>Myrtus</i> spp.                                                                                                                                                    |
| Narcissus                                           | <i>Narcissus pseudonarcissus</i>                                                                                                                                      |
| Nectarine, nonbearing                               | <i>Prunus persica</i>                                                                                                                                                 |
| Oak, bur, red                                       | <i>Quercus</i> spp. -<br><i>Q. macrocarpa</i> , <i>Q. rubra</i>                                                                                                       |
| Oleander                                            | <i>Nerium oleander</i>                                                                                                                                                |
| Olive, fragrant tea                                 | <i>Osmanthus fragrans</i>                                                                                                                                             |
| Pampas grass                                        | <i>Cortaderia</i> spp.                                                                                                                                                |
| Pansy                                               | <i>Viola</i> spp.                                                                                                                                                     |
| Peach, nonbearing                                   | <i>Prunus persica</i>                                                                                                                                                 |
| Pear, nonbearing                                    | <i>Pyrus</i> spp.                                                                                                                                                     |
| Pecan, nonbearing                                   | <i>Carya illinoensis</i>                                                                                                                                              |
| Periwinkle, Madagascar                              | <i>Catharanthus roseus</i>                                                                                                                                            |
| Periwinkle, perennial                               | <i>Vinca major</i> , <i>V. minor</i>                                                                                                                                  |
| Petunia*                                            | <i>Petunia</i> spp.                                                                                                                                                   |
| Phlox                                               | <i>Phlox</i> spp.                                                                                                                                                     |

(continued)

**Table 4. Empress® Intrinsic® brand fungicide  
Tolerant Plant Species** (continued)

| Common Name                    | Scientific Name                                                                                            |
|--------------------------------|------------------------------------------------------------------------------------------------------------|
| Pine, black, white, blue, mugo | <i>Pinus</i> spp. -<br><i>P. thunbergiana</i> ,<br><i>P. strobus</i> , <i>P. pinea</i> ,<br><i>P. mugo</i> |
| Pine, European                 | <i>Abies alba</i>                                                                                          |
| Pistachio, nonbearing          | <i>Pistacia vera</i>                                                                                       |
| Pittosporum, Japanese          | <i>Pittosporum tobira</i>                                                                                  |
| Plum, nonbearing               | <i>Prunus domestica</i>                                                                                    |
| Plum, purple leaf              | <i>Prunus cerasifera</i>                                                                                   |
| Poinsettia                     | <i>Euphorbia pulcherrima</i>                                                                               |
| Poplar                         | <i>Populus</i> spp. -<br><i>P. trichocarpa</i> ,<br><i>P. deltoides</i>                                    |
| Primrose                       | <i>Oenothera speciosa</i>                                                                                  |
| Privet                         | <i>Ligustrum</i> spp.                                                                                      |
| Purslane                       | <i>Portulaca</i> spp.                                                                                      |
| Quince                         | <i>Chaenomeles</i> spp.                                                                                    |
| Ranunculus                     | <i>Ranunculus</i> spp.                                                                                     |
| Rhaphiolepis                   | <i>Rhaphiolepis</i> spp.                                                                                   |
| Redbud                         | <i>Cercis</i> spp.                                                                                         |
| Redtip photinia                | <i>Photinia fraseri</i>                                                                                    |
| Rhododendron                   | <i>Rhododendron</i> spp.                                                                                   |
| Rock cress                     | <i>Arabis caucasica</i>                                                                                    |
| Rose                           | <i>Rosa</i> spp.                                                                                           |
| Rose mallow                    | <i>Hibiscus moscheutos</i>                                                                                 |
| Ruellia                        | <i>Ruellia</i> spp.                                                                                        |
| Russian arborvitae             | <i>Microbiota decussata</i>                                                                                |
| Sage, silverado                | <i>Leucophyllum</i> spp.                                                                                   |
| Sago                           | <i>Cycas revoluta</i>                                                                                      |
| Salvia                         | <i>Salvia coccinea</i>                                                                                     |
| Scabious, sweet                | <i>Scabiosa atropurpurea</i>                                                                               |
| Sedum                          | <i>Sedum</i> spp.                                                                                          |
| Snapdragon                     | <i>Antirrhinum</i> spp.                                                                                    |
| Speedwell                      | <i>Veronica spicata</i>                                                                                    |
| Spindle tree, burning bush     | <i>Euonymus</i> spp.                                                                                       |
| Spiraea                        | <i>Spiraea</i> spp.                                                                                        |
| Spruce                         | <i>Picea</i> spp.                                                                                          |
| Spurge, Japanese               | <i>Pachysandra terminalis</i>                                                                              |
| St. John's wort                | <i>Hypericum calycinum</i>                                                                                 |
| Stonecrop                      | <i>Sedum</i> spp.                                                                                          |
| Sweetspire                     | <i>Itea</i> spp.                                                                                           |
| Sweet William                  | <i>Dianthus barbatus</i>                                                                                   |

(continued)

**Table 4. Empress® Intrinsic® brand fungicide  
Tolerant Plant Species** (continued)

| Common Name                | Scientific Name                                            |
|----------------------------|------------------------------------------------------------|
| Thrift                     | <i>Armeria maritima</i>                                    |
| Tick seed                  | <i>Coreopsis</i> spp.                                      |
| Tulip                      | <i>Tulipa</i> spp.                                         |
| Verbena                    | <i>Verbena</i> spp.                                        |
| Viburnum, water elder      | <i>Viburnum</i> spp.                                       |
| Vinca, annual              | <i>Catharanthus roseus</i>                                 |
| Viola                      | <i>Viola</i> spp.                                          |
| Walnut tree, black, common | <i>Juglans</i> spp. -<br><i>J. nigra</i> , <i>J. regia</i> |
| Wormwood                   | <i>Artemisia</i> spp.                                      |
| Yarrow                     | <i>Achillea</i> spp.                                       |
| Zinnia                     | <i>Zinnia</i> spp.                                         |

\* **Begonia**, **impatiens** and **petunia** occasionally have shown discoloration on the flowers following **Empress** applications made directly onto the flowers. Be cautious with **Empress** applications when these species are flowering. Not all cultivars and flower colors have been evaluated. Before making **Empress** applications on the entire area, treat a small area first to ensure that a phytotoxic response will not occur.

**Table 5. Plant Species NOT Tolerant to Empress**  
**DO NOT expose these species or varieties to Empress.**

| Common Name                                                                                      | Scientific Name                |
|--------------------------------------------------------------------------------------------------|--------------------------------|
| Grape<br>Concord, Fredonia, Niagara,<br>Noiret (NY73.0136.17),<br>Rougeon, Steuben and<br>Worden | <i>Vitis</i> sp.               |
| Nine bark                                                                                        | <i>Physocarpus opulifolius</i> |
| Wintercreeper                                                                                    | <i>Euonymus vegetus</i>        |

## Conditions of Sale and Warranty

The **Directions For Use** of this product reflect the opinion of experts based on field use and tests. The directions are believed to be reliable and must be followed carefully. However, it is impossible to eliminate all risks inherently associated with the use of this product. Crop injury, ineffectiveness or other unintended consequences may result because of such factors as weather conditions, presence of other materials, or use of the product in a manner inconsistent with its labeling, all of which are beyond the control of BASF Agricultural Solutions US LLC ("BASF") or the Seller. To the extent consistent with applicable law, all such risks shall be assumed by the Buyer.

BASF warrants that this product conforms to the chemical description on the label and is reasonably fit for the purposes referred to in the **Directions For Use**, subject to the inherent risks, referred to above.

**TO THE EXTENT CONSISTENT WITH APPLICABLE LAW, BASF MAKES NO OTHER EXPRESS OR IMPLIED WARRANTY OF FITNESS OR MERCHANTABILITY OR ANY OTHER EXPRESS OR IMPLIED WARRANTY.**

**TO THE EXTENT CONSISTENT WITH APPLICABLE LAW, BUYER'S EXCLUSIVE REMEDY AND BASF'S EXCLUSIVE LIABILITY, WHETHER IN CONTRACT, TORT, NEGLIGENCE, STRICT LIABILITY, OR OTHERWISE, SHALL BE LIMITED TO REPAYMENT OF THE PURCHASE PRICE OF THE PRODUCT.**

**TO THE EXTENT CONSISTENT WITH APPLICABLE LAW, BASF AND THE SELLER DISCLAIM ANY LIABILITY FOR CONSEQUENTIAL, EXEMPLARY, SPECIAL OR INDIRECT DAMAGES RESULTING FROM THE USE OR HANDLING OF THIS PRODUCT.**

BASF and the Seller offer this product, and the Buyer and User accept it, subject to the foregoing **Conditions of Sale and Warranty** which may be varied only by agreement in writing signed by a duly authorized representative of BASF.

1108

***Empress, Intrinsic, Orvego, and Stature** are registered trademarks of BASF.*

***Maxx and Subdue** are registered trademarks of a Syngenta Group Company.*

***Segway** is a registered trademark of FMC Professional Solutions.*

© 2025 BASF Agricultural Solutions US LLC  
All rights reserved.

007969-00355.20250214.**NVA 2025-04-0384-0051**  
Based on: NVA 2023-04-0384-0079  
Supersedes: NVA 2024-04-0384-0278

BASF Agricultural Solutions US LLC  
2 TW Alexander Drive  
Research Triangle Park, NC 27713

  
We create chemistry